Module - I: Introduction to Group Theory

1 Groups : General Concepts

Let S be a set and $*$ be a *binary operation* on S: $(x, y) \mapsto x * y$ for $x, y \in S$.

Definition 1. *An element $e \in S$ is an* identity *with respect to $*$ if $x * e = e * x = x \,\forall\, x \in S$.*

Definition 2. *An element $x \in S$ has an* inverse, *denoted x^{-1}, in S w.r.t. $*$ if $x * x^{-1} = x^{-1} * x = e$. If an inverse exists, the element is termed* invertible.

An Evolution:

- *Natural Numbers:* $\mathbb{N} = 0, 1, 2, 3 \cdots$
  With the operation $+$, i.e. the natural addition, the *identity* is 0, but no element possesses inverse.
- *Integers:* $\mathbb{Z} = \cdots, -3, -2, -1, 0, 1, 2, 3 \cdots$
  Both $+$ and $\times$, integer addition and multiplication, are defined with identities $0, 1$ respectively. Inverse exists for all elements *only* for $+$. (Which elements of $\mathbb{Z}$ have multiplicative inverses?)
- *Rational Numbers:* $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$. Inverse now exists for all elements for $+$, and for all *non-zero* elements for $\times$.

Informally, $\mathbb{Z}$ is a *Group* with respect to the operation $+$, and a *Ring* having operations $+, \times$. $\mathbb{Q}$ is a *Field* with operations $+, \times$. The real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$ are fields as well.

1.1 Groups and Subgroups

Definition 3. *A* Group *is a non-empty set with a* binary operation *'$*$' which satisfies the following* axioms*:*

(i) *For $x, y \in G$, $x * y \in G$ (*Closure*);*
(ii) $(x * y) * z = x * (y * z)$ *(*Associativity*);*
(iii) *There exists $e \in G$ such that $e * x = x * e = x$ for all $x \in G$ (*Identity exists*);*
(iv) *For any $x \in G$, there exists $x^{-1} \in G$ such that $x * x^{-1} = e = x^{-1} * x$ (*Inverse exists*).*

Examples:

1. $(\mathbb{Z}, +)$ is a group, $(\mathbb{Z}, .)$ is not!
2. *Any field* is a group with respect to the additive operation, while all its non-zero elements form a multiplicative group.
3. The set of invertible matrices of a given order, say n, over a field F, forms a group under matrix multiplication. This group is termed the *General Linear Group* and often denoted by $\mathrm{GL}_n(F)$. (Can one define an additive group of matrices over any field?)
4. *Finite Groups from $\mathbb{Z}$:* Given a positive integer n, consider the set $\mathbb{Z}_n = \{[0], [1], [2], \cdots, [n-1]\}$, where $[k]$ denotes the k-th *residue class* of all integers *modulo* n, i.e. it represents all the integers which give remainder k when divided by n. This is a group under addition modulo n. For instance, $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ is a group under addition modulo 4.

In the remainder of the note, when the operation of a group G is defined *additively*, the identity is denoted by 0 and the inverse of $x \in G$ is denoted by $-x$. Likewise, for a *multiplicative* group, the identity is denoted by 1 and the inverse of $x \in G$ is x^{-1}.

Definition 4. *A non-empty subset H of a group G is termed a* subgroup *of G if H is a group with the same operation as G.*

Examples of Subgroups

- Consider $H = \{[0], [2]\} \subset \mathbb{Z}_4$. It can be shown that it is a group under addition modulo 4, and hence, a subgroup of $\mathbb{Z}_4$. Can the same be said for the subset $\{[0], [1]\}$?
- The *Special Linear Group* of matrices over a field F, denoted $SL_n(F)$ is the set of all $n \times n$ matrices with determinant 1. Verify that it is indeed a subgroup of the general linear group $GL_n(F)$.

We now present a result which outlines several equivalent ways to characterize a subgroup.

Proposition 1. *Let H be a non-empty subset of a group $(G, .)$. Then the following are equivalent (TFAE):*

1. *H is a subgroup of G.*
2. *For $x, y \in H$, both x^{-1} and xy belong to H.*
3. *For $x, y \in H$, $xy^{-1} \in H$.*

Proof. Can prove: 1. $\implies$ 3. $\implies$ 2. $\implies$ 1.
3. $\implies$ 2. : By 3., for a non-zero $x \in H$, $xx^{-1} = 1 \in H$, i.e. identity exists. This in turn implies that $1.x^{-1} = x^{-1} \in H$, again by 3.; hence, inverse exists for x. Hence $x, y \in H \implies x^{-1}, y^{-1} \in H \implies x(y^{-1})^{-1} = xy \in H$.
(Complete the other steps.)

□

Exercise: Given an integer n, consider the set $n\mathbb{Z} = \{nz : z \in \mathbb{Z}\}$, i.e. the set of integer multiples of n. Prove that $n\mathbb{Z}$ is an additive subgroup of $\mathbb{Z}$ by all the equivalent checks outlined in Proposition 1.

1.2 Cosets and Lagrange's Theorem

In this subsection, we delve further into the structure of groups via subgroups.

Definition 5. *Let $(G, +)$ be a group and H, a subgroup of G. For any $g \in G$, the set $g + H = \{g + h : h \in H\}$ is a* left coset *of H in G; similarly $H + g$ is a* right coset.

In multiplicative notation, $gH = \{gh : h \in H\}$ is a left coset of H in G and similarly, Hg is a right coset.
As an instance, the (additive) cosets of $H = \{0, 2\}$ in $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ are: $0 + H = \{0, 2\}$, $1 + H = \{1, 3\}$, $2 + H = \{2, 0\}$, $3 + H = \{3, 1\}$. It follows that there are only *two* distinct cosets and, moreover, the left and right cosets coincide by commutativity.

Lemma 1. *Let G be a group and H a subgroup of G. Then the relation $\sim$ on G, defined as: $a \sim b \iff a - b \in H$, is an equivalence relation.*

Proof. (Sketch) *Reflexive:* $a \sim a$ as $a - a \in H$. (Why?)
Symmetric: $a \sim b \implies a - b \in H \implies b - a \in H$ (Why?)
Transitive: $a \sim b, b \sim c \implies a - b = h_1 \in H, b - c = h_2 \in H \implies a - c = h_1 + h_2 \in H$.

□

Exercise: State and prove Lemma 1 in the multiplicative notation.

Lemma 2. *Given an equivalence relation $\sim$ on a set S, there exists a unique partition of S; the elements of the partition are the equivalence classes of $\sim$ and vice versa.*

Proof: Exercise[1].

If S be a set endowed with an equivalence relation $\sim$, then for an element $a \in S$, denote the equivalence class of a as the set: $[a]_\sim = \{s \in S : s \sim a\}$.

Lemma 3. *Let G be a group, H a subgroup of G, and $\sim$ be the equivalence relation defined in Lemma 1. Then for $g \in G$, $[g]_\sim$ is a left coset of H in G.*

Proof: Exercise.

Given the additive subgroup $n\mathbb{Z}$, for some integer n, of $\mathbb{Z}$, the corresponding equivalence relation is: $z_1 \sim z_2 \iff n|z_1 - z_2$ for $z_1, z_2 \in \mathbb{Z}$. There are n equivalence classes, given by: $[0], [1], \cdots, [n-1]$, where the relation subscript has been dropped.
The cosets of $n\mathbb{Z}$ in $\mathbb{Z}$ are thus given by:
$[0] = n\mathbb{Z} = \{nz : z \in \mathbb{Z}\}$; $[1] = 1 + n\mathbb{Z} = \{1 + nz : z \in \mathbb{Z}\}, \cdots$

Definition 6. *The number of (left or right) cosets of a subgroup H in a group G is called the* index *of H in G, denoted $[G : H]$.*

For instance, the index $[\mathbb{Z} : 2\mathbb{Z}] = 2$ as there are only two cosets; note that the index is finite even though both the group and its subgroup are infinite.
We now state and prove Lagrange's theorem, a result of pivotal importance in describing and counting the subgroups of a group.

Theorem 1 (Lagrange). [2] *Let H be a subgroup of a group G, of sizes $|H|, |G|$ respectively. Then*

$$|H|[G : H] = |G|.$$

In particular, if G is finite, $|H|$ divides $|G|$.

[1] For instance, cf. *Topics in Algebra*, I. N. Herstein, Chapter 1.

[2] Joseph-Louis Lagrange (1736 - 1813), born Giuseppe Lodovico Lagrangia, was an Italian mathematician of French descent. He was initially trained for a legal career, and was interested in classical languages. A treatise on astronomy sparked off an interest in mathematics, but he was largely self-taught when he started producing results that were significant enough to interest Euler. In his highly productive and distinguished career, Lagrange made profound contributions in analytical mechanics (Lagrangian mechanics), analysis (variational calculus and differential equations), algebra, number theory and astronomy. Napoleon bestowed state honours on him and made him a Count. He was a mentor to both Fourier and Poisson.

Fig. 1: Joseph Louis Lagrange's statue in Turin, Italy

Proof. By Lemmas 2 and 3, G is a disjoint union of the cosets of H. So it is enough to prove that the size of each coset equals $|H|$.
Define maps α_g and β_g between H and any (left) coset $g+H$, $g \in G$, as follows:

$$\alpha_g : H \to g + H;\ h \mapsto g + h.$$

$$\beta_g : g + H \to H;\ g + h \mapsto -g + (g + h).$$

It is easily verified that α_g, β_g are an inverses of each other. Hence $\alpha_g : H \to g + H$ is a bijection, which implies that the cardinality of *any* coset of H in G equals $|H|$. Hence the theorem.

$\square$

Definition 7. *If G is a group, the (multiplicative) order of $g \in G$, denoted $\mathrm{o}(g)$, is the least positive integer m such that ($g^m = 1$) $mg = 0$. If no such integer exists, the order of g is said to be infinite.*

The cardinality of a group G, hitherto denoted $|G|$, is often termed the *order* of the group and denoted $\mathrm{o}(G)$ as well.

Definition 8 (Cyclic Subgroups). *For $g \in G$, a multiplicative group, the set $\{1, g, g^2, \cdots\}$ is called the* cyclic subgroup *of G generated by g and denoted $< g >$.*

An analogous definition exists for additive groups.
We next collect a few results which are immediate consequences of Lagrange's theorem.

Corollary 1. *Let G be a finite, multiplicative group with $g \in G$. Then* $\mathrm{o}(g) \mid \mathrm{o}(G)$.

Proof. Let $g \in G$ have order $\mathrm{o}(g) = m$; we claim that the cyclic subgroup $< g >$ has order m.
Clearly $< g > \subseteq \{1, g, \cdots, g^{m-1}\}$, as m is the least integer which satisfies: $g^m = 1$. If $\mathrm{o}(< g >)$ is strictly less than m, there exist exponents $1 \leq j < i \leq m$ such that $g^i = g^j$, which implies $g^{i-j} = 1$. This is a contradiction, as $i - j < m$, the order of g. Hence the order of the cyclic subgroup $< g >$ of the group G is precisely m.
The assertion now follows from Lagrange's theorem.

$\square$

Corollary 2. *If G is a finite multiplicative group and a any element of G, then* $a^{\mathrm{o}(G)} = 1$.

Proof: Exercise.

Corollary 3. *If G is a finite multiplicative group with* $\mathrm{o}(G) = p$*, a prime number. Then G is a cyclic group.*

Proof. As the order of G is the prime number p, using Lagrange's theorem (Theorem 1), one can conclude that the order of every subgroup H of G is either 1 or p. This implies that the only subgroups are $H =< 1 >$ and $H = G$. So if H is the cyclic subgroup of G generated by any element $a \in G$ such that $a \neq 1$, then $H =< a >= G$. This proves the assertion.

$\square$

Exercise:

1. State and prove Lagrange's Theorem for a multiplicative group.
2. State and prove Corollaries 1-3 for additive groups.

1.3 Normal Subgroups

The notion of a normal subgroup of a group is, in one sense, related to the extent of *commutativity* of the group operation. The definition of a normal subgroups leads the way to formulating the structure of a *quotient group*. We begin with a proposition which explores commutativity at the level of subsets within a group.
In this subsection, we use the multiplicative notation.

Proposition 2. *If K, H are two subgroups of a group G, then the set*

$$HK = \{x \in G : x = hk, h \in H, k \in K\}$$

is a subgroup of G iff (if and only if) $HK = KH$.

Proof. If $HK = KH$, then for any $x = hk$ and $y = h'k'$ in HK, we have: $xy = h\underbrace{kh'}k' = h\underbrace{h''k''}k' \in HK$.
Further, we have: $x^{-1} = k^{-1}h^{-1} \in KH = HK$ and so, HK is a subgroup.
The converse is left as an **exercise**.
(*Hint:* Showing that any $x \in HK$ also $\in KH \implies HK \subseteq KH$...) ☐

Definition 9. *A subgroup N of a group G is a* normal subgroup, *denoted $N \trianglelefteq G$, if, for any $g \in G$ and $n \in N$, $gng^{-1} \in N$.*

Proposition 3. *The subgroup $N \trianglelefteq G$ if and only if:*

1. *for all $g \in G$, $gNg^{-1} = N$.*
2. *every left coset of G is a right coset.*
3. *the product of any two right (or left) cosets of N is again a right (left) coset.*

Proof. 1. If $N \trianglelefteq G$, we have $gng^{-1} \in N$ for all $g \in G$, which implies $gNg^{-1} \subset N$. Same argument yields $g^{-1}Ng \subset N$. But then we have: $N = g(g^{-1}Ng)g^{-1} \subset gNg^{-1}$, whence $N = gNg^{-1}$.
Conversely, $gNg^{-1} = N, \forall g \in G \implies gng^{-1} \in N, \forall n \in N$ and $\forall g \in G$, which, by definition, means that $N \trianglelefteq G$.
2. By 1., if $N \trianglelefteq G$, $gNg^{-1} = N \implies gN = Ng$, i.e. each left coset is a right coset.
Conversely, if every left coset is a right coset, we have: $gN = Ng'$, $g, g' \in G$. But as cosets are disjoint, it follows that $Ng' = Ng$ (as $g \in Ng'$), and so, $gN = Ng$, $\forall g \in G$.
3. To prove that $N \trianglelefteq G \implies$ the product $Ng_1Ng_2 = Ng$, for some $g \in G$, is left as an **exercise**.
Conversely, let the product of any two right cosets be a right coset, i.e. $Ng_1Ng_2 = Ng$ for some $g \in G$. Then for all $g_1 \in G$, choosing $g_2 = g_1^{-1}$ yields: $Ng_1Ng_1^{-1} = Ng$ for some $g \in G$. But $1 \in Ng_1Ng_1^{-1}$, and this $\implies 1 \in Ng \implies g^{-1} \in N \implies g \in N$,

i.e. $Ng = N$. Hence for all $g_1 \in G$, we have $Ng_1Ng_1^{-1} = N$, i.e. $n_1g_1n_2g_1^{-1} \in N$, for any $n_1, n_2 \in N$ and $\forall g_1 \in G$. This in turn implies $g_1n_2g_1^{-1} \in N$, $\forall n_2 \in N$ and $\forall g_1 \in G$. Hence the proof.

□

An important consequence of the definition of a normal subgroup is that it leads to the notion of a *quotient group*, as stated in the following theorem.

Theorem 2. *If G/N denotes the set of right (or left) cosets of $N \trianglelefteq G$, then G/N is a group, termed the quotient group of G by N.*

Proof. **Exercise**.
(*Hint:* Define the group operation $*$ as follows: $Ng_1 * Ng_2 = Ng_1g_2$. Identity and inverse elements are defined in accordance.)

Practice Suggestions:
From *Topics in Algebra*, I. N. Herstein, Chapter 2:

1. Section 5, Problems 1-5, 12.
2. Section 6, Problems 2-5, 9 -12.

2 The Group Isomorphism Theorems

In this section we explore some specific mappings between groups and among the subgroups of a group. The notions of normal subgroups and quotient groups will be frequently made use of.

2.1 Group Homomorphisms

Definition 10. *A mapping f from a group $(G, .)$ to another group $(G', *)$ is termed a (group)* homomorphism *if:*

$$\forall\, a, b \in G,\ \ f(a.b) = f(a) * f(b).$$

– Let $G = (\mathbb{Z}, +)$; define the map $\phi : G \to G$; $x \mapsto 2x$. Then ϕ is a homomorphism from G to itself: for integers $x_1, x_2 \in G$, $\phi(x_1 + x_2) = 2(x_1 + x_2) = 2x_1 + 2x_2 = \phi(x_1) + \phi(x_2)$.
– Consider $\mathbb{Z}_n$ to be the group of integers under addition modulo n. Define $\phi : \mathbb{Z} \to \mathbb{Z}_n;\ x \mapsto \overline{x}$, where $\overline{x}$ is the remainder of x on division by n, i.e. $\overline{x} = x \mod n$. It can be readily verified that this is a homomorphism as well.

Under a group homomorphism, identity and inverse elements correspond, as set down in the following

Lemma 4. *Let $\phi : G \to G'$ be a group homomorphism. Then*

i. $\phi(1_G) = 1_{G'}$ *;*
ii. For all $x \in G$, $\phi(x^{-1}) = (\phi(x))^{-1}$.

Proof. **Exercise.**

Lemma 5. *Let $N \trianglelefteq G$ and define the map:*

$$\phi : G \to G/N;\ \phi(g) = Ng\,\forall\, g \in G.$$

Then ϕ is a surjective[3] *homomorphism of G onto G/N.*

Proof. Surjectivity is obvious.
To prove homomorphism: we have, for $x, y \in G$,
$$\phi(xy) = Nxy = NxNy = \phi(x)\phi(y).$$

□

Definition 11. *Given a homomorphism $\phi : G \to G'$, the* kernel *of the homomorphism, denoted* Ker_ϕ, *is defined as the set:* $\{x \in G : \phi(x) = 1'\}$, $1'$ *being the identity element of* G'.

Lemma 6. *If $\phi : G \to G'$ is a group homomorphism with kernel* $\mathrm{Ker}_\phi = K$, *then $K \trianglelefteq G$.*

Proof. If $g, h \in K$, then $\phi(g) = \phi(h) = 1'$ and so, $\phi(gh) = \phi(g)\phi(h) = 1'$. This proves closure as a group. The remaining axioms of a group can be likewise verified.
To prove that K is, in addition, a normal subgroup, we have, for any $g \in G$ and $k \in K$, $\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g^{-1}) = \phi(g)1'[\phi(g)]^{-1} = 1'$, i.e. $gkg^{-1} \in K$. Hence the assertion.

□

[3] A function $f : A \to B$ between sets A, B is *surjective* or *onto* if, for every $b \in B$, there exists at least one $a \in A$ such that $f(a) = b$.

2.2 The Group Isomorphism Theorems

Definition 12. *A homomorphism is termed an* isomorphism *if it is bijective. Two groups G, G' are said to be isomorphic, denoted $G \simeq G'$, if there exists an isomorphism between them.*

Exercise Let $\phi : G \to G'$ be a surjective group homomorphism and let $K = \mathrm{Ker}_{\phi}$. Let $x \in G$ be *any* pre-image of $g' \in G'$, i.e. $\phi(x) = g'$. Prove that the set of pre-images of g', given by $\{g \in G : \phi(g) = g'\}$ is exactly identical with Kx.
Hence prove that ϕ is an isomorphism iff $K = \{1_G\}$.

Theorem 3. *Let $f : G \to H$ be a group homomorphism with kernel K and image $H' \subseteq H$. Then*

$$\sigma : G/K \to H';\ \overline{a} \mapsto f(a)$$

where $\overline{a}$ is the image of $a \in G$ in the quotient group G/K, is a group isomorphism, i.e. $G/K \simeq H'$.

Proof. (*Sketch*)

- σ is a group homomorphism:

$$\sigma(\overline{a} + \overline{b}) = \sigma(\overline{a + b}) = f(a + b) = f(a) + f(b)$$

- Surjectivity of σ follows from definition.
- Injectivity: $\sigma(\overline{a}) = \sigma(\overline{b}) \implies f(a) = f(b) \implies a - b \in K$.

$\square$

Definition 13. *For any non-empty subset A of a group G and $g \in G$, define the set $gAg^{-1} := \{gag^{-1} : a \in A\}$. The* normalizer *of the set A in G is defined as the set:*

$$N_G(A) = \{g \in G : gAg^{-1} = A\}$$

Theorem 4. *Let G be a group and A, B subgroups of G such that $A \subseteq N_G(B)$. Then AB is a subgroup of G and $A \cap B \trianglelefteq A$. Further, there exists an isomorphism of quotient groups: $AB/B \simeq A/(A \cap B)$.*

Proof. To show that AB is a subgroup of G, enough to show that $AB = BA$ (by Proposition 2). As $A \subseteq N_G(B)$, for any $a \in A$ and $b \in B$, we have: $aba^{-1} \in B$, whence $ab = b'a$ for some $b' \in B$. Thus $AB \subseteq BA$. It can be shown similarly that $BA \subseteq AB$. Hence the first assertion.

Consider $g \in A \cap B$ and $a \in A$; obviously, $aga^{-1} \in A$. But, by hypothesis, $aga^{-1} \in B$, as $A \subseteq N_G(B)$. Therefore, $aga^{-1} \in A \cap B$, which implies that $A \cap B \trianglelefteq A$.

To prove that $AB/B \simeq A/(A \cap B)$, we first define a homomorphism

$$\phi : AB \to A/(A \cap B);\ ab \mapsto \overline{a}$$

where $\overline{a}$ is the image of $a \in A$ in the quotient group $A/(A \cap B)$ (Verify that $A/(A \cap B)$ is indeed a group!) and invoke Theorem 3 using the kernel of this homomorphism.
Claim: The map ϕ is a surjective homomorphism.
We have for arbitrary $ab, a_1b_1 \in AB$,

$$aba_1b_1 = aa_1(a_1^{-1}ba_1)b_1 = aa_1b'b_1$$

where $b' = (a_1^{-1}ba_1) \in B$, as $A \in N_G(B)$.
It follows that

$$\phi(aba_1b_1) = \phi(aa_1b'b_1) = \overline{aa_1} = (\overline{a})(\overline{a_1}) = \phi(ab)\phi(a_1b_1).$$

The surjectivity of this homomorphism is easily checked.

By Theorem 3, enough to prove that $\mathrm{Ker}_\phi = B$.
To that end, for any $b \in B \subseteq AB$, we have: $\phi(b) = \phi(1_Ab) = \overline{1_A}$ which implies that $B \subseteq \mathrm{Ker}_\phi$.
Again for any $x \in \mathrm{Ker}_\phi$, $x = ab$ such that $\overline{a} = \overline{1_A} \implies a \in A \cap B$. It follows that $x \in B$ which implies $\mathrm{Ker}_\phi \subseteq B$.
Hence the theorem.

$\square$

Theorem 5. *If H, K are normal subgroups of a group G such that $H \subseteq K$, then*

$$(G/H)/(K/H) \simeq G/K.$$

Proof. (*Sketch*)

- The map $\sigma : G \to G/K;\ g \mapsto gK$, taking an element in G to the corresponding coset of K in G, is a surjective group homomorphism with kernel K.
- The map $\overline{\sigma} : G/H \to G/K;\ gH \mapsto gK$, between the cosets of H and K in G, respectively, is again a surjective group homomorphism.
- The kernel of $\overline{\sigma}$ is K/H; using Theorem 3 yields the assertion.

Miscellaneous Problems

1. Prove that the special linear group $SL_n(\mathbb{R})$ is a normal subgroup of the general linear group $GL_n(\mathbb{R})$ (cf. Section 1).
2. Let N, M be two normal subgroups of an arbitrary group G such that $N \cap M = \{\mathrm{id}_G\}$. Prove that for any $n \in N$ and $m \in M$, $nm = mn$.
3. Given a group G, consider the subset $U = \{xyx^{-1}y^{-1} : x, y \in G\}$. The smallest group U_G which contains the set U is called the *commutator subgroup* of G. Prove that $U_G \trianglelefteq G$.
4. If G is the group of non-zero real numbers under multiplication, determine if the following maps are homomorphisms. Also determine the kernels if they are.
   i. $\phi : G \to G;\ x \mapsto x^2$.
   ii. $\sigma : G \to G;\ x \mapsto 2^x$.
5. If G is the group of real numbers under addition, determine if $\phi : G \to G;\ x \mapsto x + 1$ is a homomorphism.
6. Fix an element $g \in G$ to define the map $\phi_g : G \to G;\ x \mapsto gxg^{-1}$. Prove that this is an isomorphism of G onto G, also termed an *automorphism* of G.
7. Show that the set $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$ is a group under coordinate-wise addition: $(x, y)+(x_1, y_1) = (x+x_1, y+y_1)$. Define the map $\phi : \mathbb{R}^2 \to \mathbb{R};\ (x, y) \mapsto y$. Determine if this map is a homomorphism and compute its kernel if it is.